AI Image Generator

9. Operations and Troubleshooting

Version 1.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

9. Operations and Troubleshooting

9.1 Starting and stopping

Action	How
Start	Double-click run-app.bat, or `.\.venv\Scripts\python.exe app.py` in the project folder
Confirm ready	Console shows "Model ready in N s", "Upscaler ready", "Running on local URL: http://127.0.0.1:7860"
Stop	Ctrl+C in the console, or close the window
Change port	Edit the server_port value in the last line of app.py
Expose on the LAN	Change server_name to "0.0.0.0" in app.py. There is no authentication, so only do this on a trusted network

9.2 Logs

The app prints one line per generated image to the console, for example:

```
#4 saved 20261022-005957_seed42.png  1344x768  13.527s  peak 10.46 GB  52.0 C  -> 3840x2194 in 3.38s
```

To keep a log file, redirect: `.\.venv\Scripts\python.exe -u app.py > app.log 2>&1`.

9.3 Known issues encountered during development and their fixes

Symptom	Cause	Fix applied
AttributeError: 'StableDiffusionXLPipeline' object has no attribute 'enable_vae_slicing'	diffusers 0.41 removed the pipeline-level helper	Call pipe.vae.enable_slicing() instead
Parser ignored "seed 42" and misread "21:9 steps 45" as 9 steps	Directive removal used offsets from a stale lowercase copy of the prompt	Rewrote removal to search and cut the live prompt string each time
Regex word boundaries silently vanished	A patch script wrote "\b" through a normal Python string, turning it into a backspace character	Restored with a byte-level replace; patches now use raw strings or the Write tool
UnicodeEncodeError: 'charmap' codec can't encode character	Windows console code page cp1252 cannot print the arrow and dot characters in reports	Set PYTHONIOENCODING=utf-8 when printing reports from scripts; the browser is unaffected
torchvision installed as the CPU build	pip pulled torchvision from PyPI as a dependency of spandrel before the CUDA index was consulted	Reinstalled with --force-reinstall from the cu126 index; requirements.txt documents the order
Warning: NumPy array is not writable	Tiles were passed as read-only views	Copy the array before converting to a tensor
Server died when a wait loop was cancelled	The server was a child of the same shell command	Launch the server as its own background process

9.4 Troubleshooting guide

Problem	Check	Resolution
"CUDA GPU not available" at startup	`python -c "import torch; print(torch.version.cuda)"` prints None	Reinstall torch from the cu126 index inside the virtual environment
Port 7860 already in use	Task Manager shows a python.exe from a previous run	End that process, or change the port
Out of memory during generation	Other GPU programs running; or a card under 12 GB	Close other GPU apps; use generate.py which offloads to CPU on small cards
First start hangs on "Loading"	Model download in progress	Wait; check %USERPROFILE%\.cache\huggingface\hub growth
Telemetry fields show None	nvidia-smi not on PATH	Reinstall the NVIDIA driver, or add C:\Windows\System32 to PATH
Upscaler weights missing	FileNotFoundError naming models/RealESRGAN_x4plus.pth	Download from the Real-ESRGAN v0.1.0 release into models/
History tab empty	New database, or generations.db moved	Generate an image; or restore the file next to db.py
Browser slow to show an 8K result	30 MB PNG transfer	Expected; open the file from outputs/ directly

9.5 Maintenance

- Outputs grow by 1.5 MB per native image and 8 to 31 MB per upscaled image. Prune outputs/ periodically; the database keeps the path and the History tab will show "not found" for deleted files.
- Back up outputs/ and generations.db together so paths stay valid.
- To reset history, stop the app and delete generations.db; it is recreated empty on next start.
- To update dependencies, upgrade inside the virtual environment and rerun the parser self-test and one generation. Pin versions in requirements.txt if a breaking change is found.

9.6 Security notes

- The server listens on localhost only by default and has no authentication.
- No telemetry or prompts are sent anywhere; the only network access is the one-time model download from Hugging Face.
- Prompts are stored verbatim in generations.db. Treat the file with the same care as the images.